

**Analog IC Design (Xschem, Ngspice, ADT)****Lab 03****Cascode Amplifier****Intended Learning Objectives**

In this lab you will:

- Learn how to use the Sizing Assistant (SA) to size the transistors.
- Design and simulate a cascode amplifier.
- Design a bias circuit for the cascode amplifier.
- Investigate the gain, the bandwidth, and the GBW of a cascode amplifier.

**Part 1: Device Sizing Using SA**

1) From the square law, we have

$$g_m = \frac{2I_D}{V_{ov}} \rightarrow V_{ov} = \frac{2}{g_m/I_D}$$

For a real MOSFET, if we compute  $V_{ov}$  and  $\frac{2}{g_m/I_D}$  they will not be equal. We can define V-star ( $V^*$ ), which is calculated from actual simulation data using the formula

$$V^* = \frac{2}{g_m/I_D} \leftrightarrow g_m = \frac{2I_D}{V^*}$$

The lower the  $V^*$  the higher the  $g_m$ , but the larger the area and the lower the speed. An often used sweet-spot that provides good compromise between different trade-offs is  $V^* = 200mV$ .

2) Although the  $V^*$  is a nice parameter that is inspired by the square-law, it does not have an intuitive or a physical meaning (it is not an actual voltage in the circuit). We actually defined  $V^*$  in order to be able to define a relation between the  $g_m$  and  $I_D$ . Thus, the real parameter that we should care about is the  $g_m$  over  $I_D$  ratio ( $g_m/I_D$ ).

If the square-law is valid

$$g_m = \frac{2I_D}{V_{ov}} \rightarrow \frac{g_m}{I_D} = \frac{2}{V_{ov}}$$

Using  $V^*$

$$\frac{g_m}{I_D} = \frac{2}{V^*}$$

A small  $g_m/I_D$  means large  $V_{ov}$  (biasing in strong inversion) and a large  $g_m/I_D$  means small  $V_{ov}$  (biasing in weak inversion).

3) There are many good things about using the  $g_m/I_D$  as a design knob:

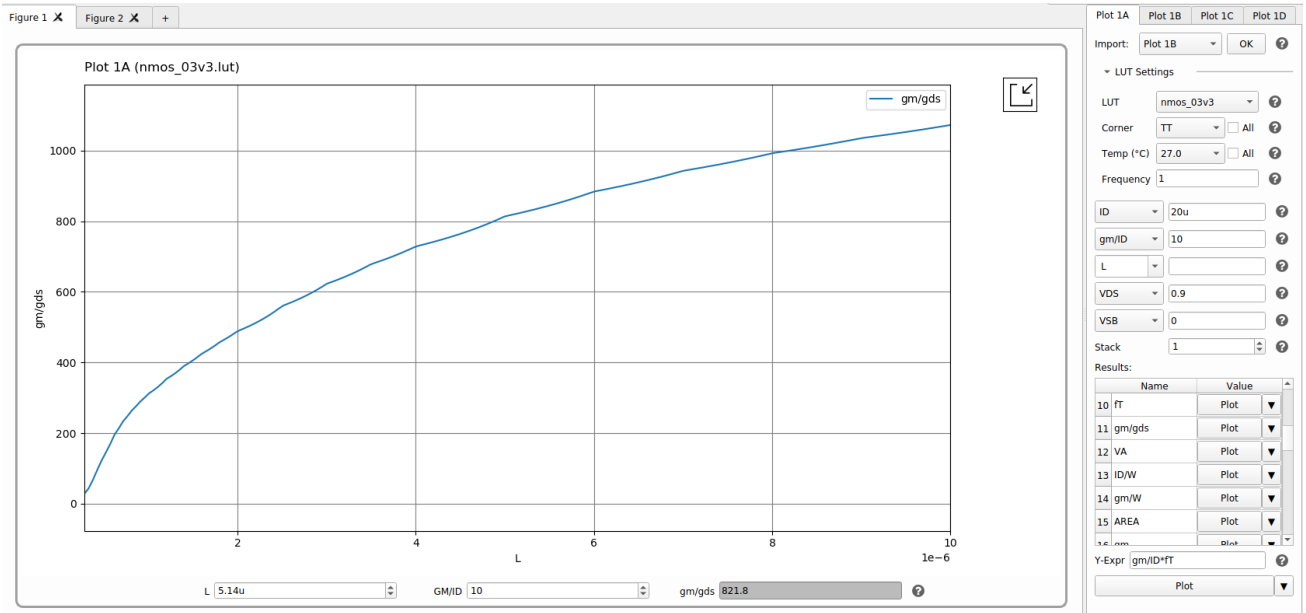
- The  $g_m/I_D$  gives a direct relation between the most important MOSFET parameter ( $g_m$ ) and the most valuable resource ( $I_D$ ). For example, a  $g_m/I_D = 10 \text{ S/A}$  means you get  $10 \mu\text{S}$  of  $g_m$  for every  $1 \mu\text{A}$  of bias current.
  - The  $g_m/I_D$  is a normalized knob: it has a limited search range (typically from 5 to 25 S/A) independent of the technology or the device type.
  - The  $g_m/I_D$  is intuitive because it tells you directly about the inversion level (bias point) and consequently all related trade-offs. For example,  $g_m/I_D = 5 \text{ S/A}$  means strong inversion (SI),  $g_m/I_D = 15 \text{ S/A}$  means moderate inversion (MI), and  $g_m/I_D = 25 \text{ S/A}$  means weak inversion (WI).
  - The  $g_m/I_D$  is an orthogonal knob: If we define the  $g_m/I_D$  then we define the inversion level (bias point). If you change  $I_D$  or  $L$  while keeping  $g_m/I_D$  fixed, then the inversion level (bias point) is kept fixed. The  $W$  is treated as an output variable instead of being treated as an input variable.
  - The higher the  $g_m/I_D$  (the lower the  $V^*$ ) the higher the efficiency and the headroom (the available swing), but the larger the area and the lower the speed. An often used sweet-spot that provides good compromise between different trade-offs is  $g_m/I_D = 10 \text{ S/A}$  ( $V^* = 200\text{mV}$ ). Larger  $g_m/I_D$  is usually used for low-voltage and low-power designs.
- 4) We want to design a common source (CS) amplifier that has ideal current source load with the following parameters<sup>1</sup>.

| Parameter                     | Value                      |
|-------------------------------|----------------------------|
| $A_v = g_m r_o$               | 50                         |
| $g_m/I_D$                     | $10 \text{ S/A}$           |
| Supply ( $V_{DD}$ )           | $1.8 \text{ V}$            |
| Quiescent (DC) output voltage | $V_{DD}/2 = 0.9 \text{ V}$ |
| Bias Current                  | $20 \mu\text{A}$           |

- 5) Since the square-law is not accurate, we cannot use it to calculate the sizing. Instead, we will use the Sizing Assistant (SA) which is a powerful analog calculator that uses LUTs that are pre-generated from the simulations. Sweep  $L$  and plot  $g_m/g_{ds}$ . Pick  $L$  using the magic cursor.

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<sup>1</sup> A relatively short  $L$  means  $C_{gd}$  is not negligible compared to  $C_{gs}$ . This will make Part 3 results more meaningful. That's one reason why a moderate  $g_m \cdot r_o$  is used in the specs.

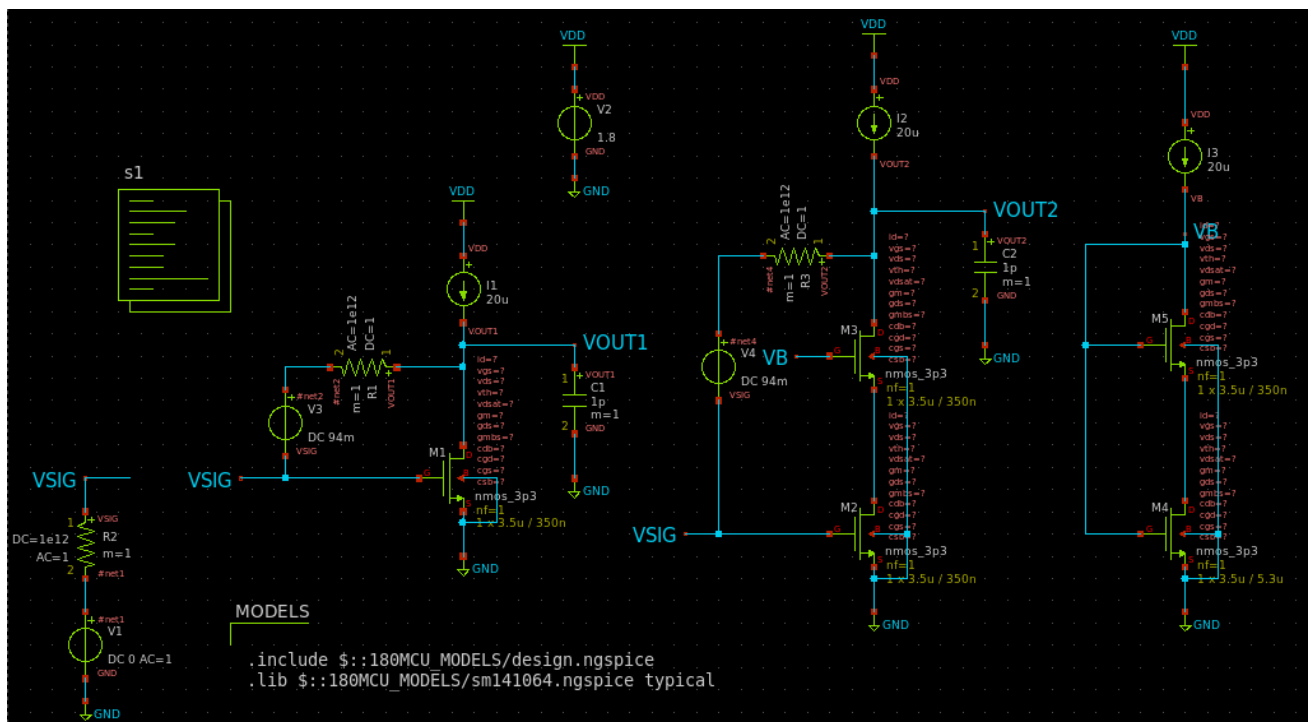


- 6) Fill the picked L back into SA to get W.
- 7) Notice that gm/gds depends on VDS. For the picked L, plot gm/gds vs VDS=0:0.9.
- 8) For the cascode amplifier, we will use the same sizing, but the VDS will be split across two transistors. What is the expected gm/gds in this case?

## Part 2: Cascode for Gain

### 1. OP Analysis

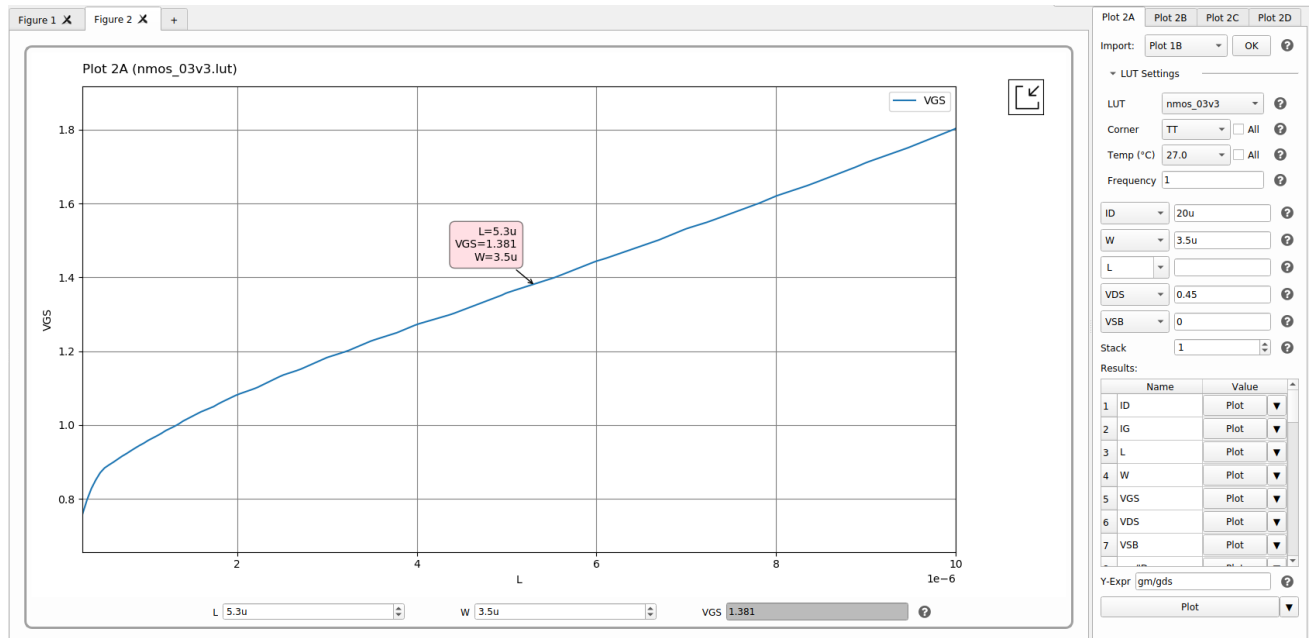
- 1) Create a new schematic. Construct the circuit shown below. Use  $I_B = 20\mu A$ . Use L and W as selected in Part 1 for M1, M2, M3, and M4. Use the same W for M4 but it will have different L as will be shown later. Use  $C_L = 1pF$ .



- 2) We need to set the quiescent (DC) output voltage of the amplifiers to bias the transistors in saturation. However, the output node is a high impedance node; thus, it is difficult to control its DC voltage.
- 3) As a simulation workaround, we use `res_ac` with different resistance values in DC/AC analysis so that it acts as s.c. or o.c. depending on the simulation type. At DC, the transistor is diode connected and  $V_{GS}$  is set by the current source. At AC, the diode connection is removed, and the input signal source is applied.
- 4) For the cascode amplifier, we will design  $V_B$  to set  $V_{DS2} \approx V_{DS3} \approx 0.45\text{ V}$  as will be shown shortly.
- 5) A dc shift (V1 and V2) is used to set the quiescent (DC) output voltage “roughly” to the required value. Setting the output voltage to exactly  $0.9\text{ V}$  is neither practical nor required.  $V_{GS0}$  and  $V_{GS1}$  can be retrieved from SA. Note that  $V_{DS1} \approx 0.9\text{ V}$  but  $V_{DS2} \approx 0.45\text{ V}$  so  $V_{GS1}$  and  $V_{GS2}$  will be slightly different. This can be ignored as  $V_{GS}$  is already always affected by  $V_{TH}$  variations, so a precise value is neither practical nor required.
- 6) To calculate  $V_B$  we need to find  $V_{GS3}$  because  $V_B = V_{GS3} + V_{DS2}$ . Note that **M3** experiences body effect, so its  $V_{GS}$  will be higher than M1 and M2.

LUT: nmos\_03v3 ?  
 Corner: TT ☐ All ?  
 Temp (°C): 27.0 ☐ All ?  
 Frequency: 1 ?  
 ID: 20u ?  
 W: [color bar] ?  
 L: [color bar] ?  
 VDS: 0.45 ?  
 VSB: 0.45 ?  
 Stack: 1 ?

- 7) M4 and M5 form **the magic battery** that will be used to generate the cascode bias voltage. Note that M5 is always in saturation (diode connected) and **M4** is always in triode (why?). We need to find the  $L$  of M4, so we set a sweep for M4 as shown below. Plot  $V_{GS}$  and find  $L$  that gives the required  $V_{GS} = V_B = V_{GS3} + V_{DS2}$ .  
 ➔ Note: To get higher VGS we need to keep  $W$  fixed and increase  $L$  or keep  $L$  fixed and decrease  $W$  to get higher VGS. Sometimes we have to do both (pick longer  $L$  and smaller  $W$ ).



- 8) Simulate the DC OP point of the above CS and cascode amplifiers. Report a snapshot showing the following parameters for M1 to M5 in addition to DC node voltages clearly annotated.

|       |
|-------|
| ID    |
| VGS   |
| VDS   |
| VTH   |
| VDSAT |
| GM    |
| GDS   |
| GMB   |
| CDB   |
| CGD   |
| CGS   |
| CSB   |

**NOTE:** “vdsat” is the minimum drain-source voltage required to bias the transistor in saturation. It is equal to  $V_{ov}$  for a square-law device. It is also referred to as “vdss” (drain-source saturation voltage) in some models. It is considered an ambiguous parameter because the transition from triode to saturation is gradual, not abrupt.

- ➔ Note: The code below shows how to save the OP parameters of every transistor in a loop, courtesy of Eng. Sheriff Shamardn. You just need to number your transistors as M1, M2, ... and set x stop value in the loop according to your design.

```
.control
```

```
alterparam sw_stat_global = 0
alterparam sw_stat_mismatch = 0
reset
```

```
save all
```

```
let x = 1
```

```

set num = {$&x}
dowhile x <= 5
save @m.xm{$num}.m0[id]
save @m.xm{$num}.m0[vgs]
save @m.xm{$num}.m0[vds]
save @m.xm{$num}.m0[vdsat]
save @m.xm{$num}.m0[vth]
save @m.xm{$num}.m0[gm]
save @m.xm{$num}.m0[gds]
save @m.xm{$num}.m0[gmb]
save @m.xm{$num}.m0[cgs]
save @m.xm{$num}.m0[cgd]
save @m.xm{$num}.m0[csb]
save @m.xm{$num}.m0[cdb]
let x = x + 1
set num = {$&x}
end

op
write lab_03_cascode_for_gain.raw
set appendwrite
ac dec 10 1 10g
remzerovec
write lab_03_cascode_for_gain.raw
.endc

```

➔ Hint: An alternative way is to print the OP parameters of all transistors in ngspice terminal (save command is NOT required). Example (you can add additional OP parameters):

```
show m : id : gm : gds : gmb : vgs : vth : vds
```

9) Check that all transistors operate in saturation. Does any transistor operate in triode? Why?

10) Do all transistors have the same  $v_{th}$ ? Why?

11) What is the relation ( $\ll$ ,  $<$ ,  $\approx$ ,  $>$ ,  $\gg$ ) between  $g_m$  and  $g_{ds}$ ?

**NOTE: use  $\gg$  or  $\ll$  if the difference is 10 times or more (one order of magnitude).**

12) What is the relation ( $\ll$ ,  $<$ ,  $\approx$ ,  $>$ ,  $\gg$ ) between  $g_m$  and  $g_{mb}$ ?

13) What is the relation ( $\ll$ ,  $<$ ,  $\approx$ ,  $>$ ,  $\gg$ ) between  $c_{gs}$  and  $c_{gd}$ ?

14) What is the relation ( $\ll$ ,  $<$ ,  $\approx$ ,  $>$ ,  $\gg$ ) between  $c_{sb}$  and  $c_{db}$ ?

## 2. AC Analysis

1) Perform AC analysis (1Hz:10GHz, logarithmic, 10points/decade) to simulate gain and bandwidth.

2) Use measure expressions to calculate parameters (DC gain, BW, GBW, and UGF) and export them to a text file.

➔ Hint: The code below shows an example of calculating the AC specs:

```

meas ac DC_GAIN_1 find vmag(vout1) at=1
let A3db = DC_GAIN_1/sqrt(2)
meas ac BW_1 when vmag(vout1)=A3db fall=1
let GBW_1 = DC_GAIN_1 * BW_1

```

```

meas ac UGF_1 when vmag(vout1)=1 fall=1

meas ac DC_GAIN_2 find vmag(vout2) at=1
let A3db = DC_GAIN_2/sqrt(2)
meas ac BW_2 when vmag(vout2)=A3db fall=1
let GBW_2 = DC_GAIN_2 * BW_2
meas ac UGF_2 when vmag(vout2)=1 fall=1

print DC_GAIN_1 BW_1 GBW_1 UGF_1 > lab_03_results.txt
print DC_GAIN_2 BW_2 GBW_2 UGF_2 >> lab_03_results.txt

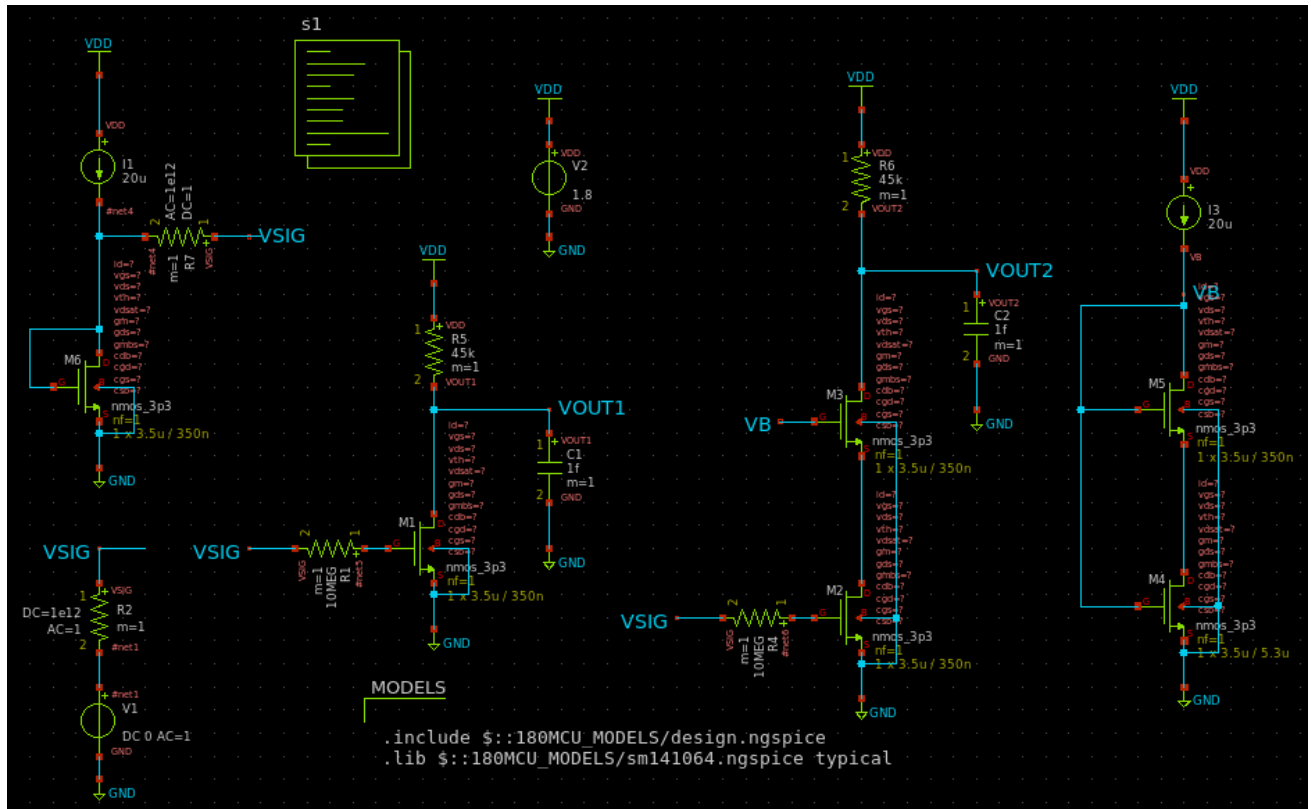
```

- 3) Report the Bode plot (magnitude) of CS and cascode **appended on the same plot**.
- 4) Using small signal parameters from OP simulation or SA, perform hand analysis to calculate DC gain, BW, and GBW of both circuits.
- 5) Report a table comparing the DC gain, BW, UGF, and GBW of both circuits from simulation and hand analysis.
- 6) Comment on the results.

## Part 3: Cascode for BW

### 1. OP Analysis

- 1) Copy the previous schematic file to a new one. Make the following modifications:
  - Remove the feedback connection used to set the DC output voltage. The DC output voltage is going to be set by the voltage drop on the resistance.
  - Replace the current source with a resistor load  $R_D$ .
  - Create a diode connected transistor (M6) that is used to generate the DC bias input voltage of the two amplifiers (replica biasing). This voltage is connected to the amplifier in DC only. In AC analysis, the AC input source is connected.
  - Set  $C_L = 1fF$  and the signal source resistance  $R_{sig} = 10M\Omega$ . This will make the dominant pole the input pole instead of the output pole.



- 2) Calculate  $R_D$  analytically such that the voltage drop on it is  $\approx V_{DD}/2$  (the current remains roughly the same as in Part 2 because we are using the VGS generated by replica biasing). Note that the DC voltage of the output node is set by the resistance ( $R_D$ ); thus, we don't need a feedback loop as in the previous case.
- 3) Simulate the DC OP point of the new CS and cascode amplifiers. Report a snapshot showing the OP parameters for M1 to M5 in addition to DC node voltages clearly annotated.
- 4) Check that all transistors operate in saturation. Does any transistor operate in triode? Why?

### AC Analysis

- 1) Perform AC analysis (1Hz:10GHz, logarithmic, 10points/decade) to simulate gain and bandwidth.
- 2) Use measure expressions to calculate parameters (DC gain, BW, GBW, and UGF) and export them to a text file.
- 3) Report the Bode plot (magnitude) of CS and cascode **appended on the same plot**.
- 4) Using small signal parameters from OP simulation or SA, perform hand analysis to calculate DC gain, BW, and GBW of both circuits.
- 5) Report a table comparing the DC gain, BW, UGF, and GBW of both circuits from simulation and hand analysis. Comment on the results.

### → Example ngspice control block:

```
.control

alterparam sw_stat_global = 0
alterparam sw_stat_mismatch = 0
reset

save all
```



```

let x = 1
set num = {$&x}
dowhile x <= 5
save @m.xm{$num}.m0[id]
save @m.xm{$num}.m0[vgs]
save @m.xm{$num}.m0[vds]
save @m.xm{$num}.m0[vdsat]
save @m.xm{$num}.m0[vth]
save @m.xm{$num}.m0[gm]
save @m.xm{$num}.m0[gds]
save @m.xm{$num}.m0[gmb]
save @m.xm{$num}.m0[cgg]
save @m.xm{$num}.m0[cdd]
save @m.xm{$num}.m0[css]
let x = x + 1
set num = {$&x}
end

op
write lab_03_cascode_for_BW.raw
set appendwrite
ac dec 10 1 10g

meas ac DC_GAIN_1 find vmag(vout1) at=1
let A3db = DC_GAIN_1/sqrt(2)
meas ac BW_1 when vmag(vout1)=A3db fall=1
let GBW_1 = DC_GAIN_1 * BW_1
meas ac UGF_1 when vmag(vout1)=1 fall=1

meas ac DC_GAIN_2 find vmag(vout2) at=1
let A3db = DC_GAIN_2/sqrt(2)
meas ac BW_2 when vmag(vout2)=A3db fall=1
let GBW_2 = DC_GAIN_2 * BW_2
meas ac UGF_2 when vmag(vout2)=1 fall=1

print DC_GAIN_1 BW_1 GBW_1 UGF_1 > lab_03_results.txt
print DC_GAIN_2 BW_2 GBW_2 UGF_2 >> lab_03_results.txt

remzerovec
write lab_03_cascode_for_BW.raw
.endc

```

## Lab Summary

In Part 1 you learned:

- How to find transistor sizing using the Sizing Assistant (SA).
- How to design a common-source and a cascode amplifier.

In Part 2 you learned:

- How to build a testbench for a cascode amplifier with current-source load.
- How to use cascode to boost the amplifier's gain.
- How to simulate the gain, the bandwidth and the GBW of a cascode amplifier with current-source load.

In Part 3 you learned:

- How to build a testbench for a cascode amplifier with resistive load.
- How to use cascode to boost the amplifier's bandwidth.
- How to simulate the gain, the bandwidth and the GBW of a cascode amplifier with resistive load.

## Acknowledgements

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